

Assignment 10 Solutions

$$10.4 \quad A - 4I = \begin{bmatrix} 1 & -6 & 2 \\ 1 & -6 & 2 \\ -1 & 6 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -6 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = 6x_2 - 2x_3 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{array} \quad x_2 \begin{bmatrix} 6 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\text{Nul}(A - 4I) = \text{span} \left\{ \begin{bmatrix} 6 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$10.11 \quad \begin{bmatrix} -1 & 1 & -1 \\ -3 & 3 & -3 \\ 1 & -3 & -2 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & -1 \\ 0 & 6 & 0 \\ 0 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & -2 & -3 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad \det(A) = \frac{(-1)^0}{(1/6)} (-1)(1)(-3) = 18$$

$$\begin{aligned} 11.1 \quad \det(A - \lambda I) &= (9 - \lambda)(-6 - \lambda) - 4(-14) \\ &= \lambda^2 - 9\lambda + 6\lambda - 54 + 56 \\ &= \lambda^2 - 3\lambda + 2 \\ &= (\lambda - 2)(\lambda - 1) \end{aligned}$$

$$\lambda = 2, 1$$

$$A - 2I = \begin{bmatrix} 7 & 4 \\ -14 & -8 \end{bmatrix} \sim \begin{bmatrix} 1 & 4/7 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -4/7 x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_1 = \begin{bmatrix} -4 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} 9 & 4 \\ -14 & -6 \end{bmatrix} + \begin{bmatrix} -4 \\ 7 \end{bmatrix} = \begin{bmatrix} -36 + 28 \\ 56 - 42 \end{bmatrix} = \begin{bmatrix} -8 \\ 14 \end{bmatrix} \checkmark$$

$$A - I = \begin{bmatrix} 8 & 4 \\ -14 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = -1/2 x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 9 & 4 \\ -14 & -6 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -9 + 8 \\ 14 - 12 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \checkmark$$

$$P = \begin{bmatrix} -4 & -1 \\ 7 & 2 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
 11.2 \quad \det(A - \lambda I) &= (2 - \lambda)(-5 - \lambda) + 12 \\
 &= \lambda^2 - 2\lambda + 5\lambda - 10 + 12 \\
 &= \lambda^2 + 3\lambda + 2 \\
 &= (\lambda + 1)(\lambda + 2)
 \end{aligned}$$

$$\lambda = -1, -2$$

$$\det(A + I) = \begin{bmatrix} 3 & -6 \\ 2 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = 2x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\det(A + 2I) = \begin{bmatrix} 4 & -6 \\ 2 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -\frac{3}{2} \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = \frac{3}{2}x_2 \\ x_2 \text{ is free} \end{array} \quad \vec{v}_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \quad D = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$10.19 \quad \text{False} \quad A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \det(A - \lambda I) = \lambda^2 + 1, \text{ no real roots}$$

$$10.20 \quad \text{False} \quad \text{Immediate from 10.19}$$

$$10.25 \quad \text{False} \quad A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix} \quad \det(A - \lambda I) = (\lambda - 3)^3$$

$$A - 3I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \dim \text{Nul}(A - 3I) = 2$$

$$10.28 \quad \text{True} \quad \det(A^T A) = \det(A^T) \det(A) = \det(A)^2 \geq 0$$

$$\begin{aligned}
 11.10 \quad \text{True} \quad &\text{We stated this. } \det(PBP^{-1} - \lambda I) = \\
 &\det(P(B - \lambda I)P^{-1}) = \\
 &\cancel{\det(P)} \det(B - \lambda I) \cancel{\det(P^{-1})} = \\
 &\det(B - \lambda I)
 \end{aligned}$$

11.11 False By 11.1, $\begin{bmatrix} 9 & 4 \\ -14 & -6 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $\vec{e}_1, \vec{e}_2$ are not eigenvectors of the left matrix

10.39 (a) $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ $\det(A) = -1$ $\det(B) = -1$
 $\det(A+B) = 0 \neq -2 = \det(A) + \det(B)$

(b) $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $\det(A-I) = (\lambda-1)^2$ $\dim \text{Nul}(A-I) = 1$

(c) $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ A is not diagonalizable

10.40 (a) $\dim \text{Nul}(A) = \dim \text{Nul}(A - 0I) \geq 1$ since 0 is an eigenvalue
 $\dim \text{Nul}(A) \leq 1$ since multiplicity of 0 in $\det(A - \lambda I)$ is 1.
so $\dim \text{Nul}(A) = 1$ and $\text{rank}(A) = 5 - \dim \text{Nul}(A) = 4$

(b) $\dim \text{Nul}(A - 2I) = 5 - \text{rank}(A - 2I) = 2$
 $\dim \text{Nul}(A + I) = 5 - \text{rank}(A + I) = 2$
 $\dim \text{Nul}(A) = 1$

$$\begin{aligned}\text{Nul}(A) &= \text{span} \{ \vec{b}_1 \} \\ \text{Nul}(A + I) &= \text{span} \{ \vec{b}_2, \vec{b}_3 \} \\ \text{Nul}(A - 2I) &= \text{span} \{ \vec{b}_4, \vec{b}_5 \}\end{aligned}$$

$\{ \vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_5 \}$ is a basis for $\mathbb{R}^5$

(a)

$$11.15 \quad \det(A - \lambda I) = (1 - \lambda)((1 - \lambda)(3 - \lambda) + 1) \\ = (1 - \lambda)(\lambda^2 - 4\lambda + 4) \\ = (1 - \lambda)(\lambda - 2)^2$$

$$\lambda = 1, 2$$

(b)

$$A - I = \begin{bmatrix} 0 & 1 & 4 \\ 0 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 \text{ is free} \\ x_2 = 0 \\ x_3 = 0 \end{array}$$

$$\text{Nul}(A - I) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$A - 2I = \begin{bmatrix} -1 & 1 & 4 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} x_1 = 3x_3 \\ x_2 = -x_3 \\ x_3 \text{ is free} \end{array}$$

$$\text{Nul}(A - 2I) = \text{span} \left\{ \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} \right\}$$

(c) Not diagonalizable, does not have eigenbasis